

KB-Spaces Characterize Monotone Stability of Sequentially Un-Compact Operators from c_0

Automated Math Discovery Run `fa_banach_001`

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Status. Candidate full solution, likely valid pending human review. The novelty search described below found no later explicit answer, but novelty confidence is moderate because the proof is short and non-arXiv literature was not exhaustively citation-indexed.

Source and open question

Kandić, Marabeh, and Troitsky define a bounded operator into a Banach lattice to be *sequentially un-compact* when the image of every bounded sequence has an un-convergent subsequence. In Proposition 9.6 of [1], they prove that the following property forces X to be a KB-space:

if $T_n \in L(c_0, X)$ are sequentially un-compact and $T_n \uparrow T$, then T is sequentially un-compact.

Immediately afterward they ask whether the converse is true. The exact source location is page 30.

Proposition 9.6. *Suppose that for every sequence (T_n) of sequentially un-compact operators in $L(c_0, X)$, $T_n \uparrow T$ implies that T is sequentially un-compact. Then X is a KB-space.*

Proof. Suppose not. Then there is a lattice isomorphism $T: c_0 \rightarrow X$. Put $x_n = Te_n$, where (e_n) is the standard unit basis of c_0 . Put $T_n = TP_n$, where P_n is the n -th basis projection on c_0 , i.e., $T_n h = \sum_{i=1}^n h_i x_i$ for $h \in c_0$. It follows that $T_n h \xrightarrow{\|\cdot\|} Th$, so that $T_n h \uparrow Th$ for every $h \geq 0$ and, therefore, $T_n \uparrow T$. For each n , T_n has finite rank and, therefore, is sequentially un-compact.

We claim that, nevertheless, T is not sequentially un-compact. Put $w_n = e_1 + \cdots + e_n$ in c_0 . Note that (w_n) is norm bounded and $Tw_n = x_1 + \cdots + x_n$. Since T is an isomorphism, (Tw_n) is not norm-convergent. Since (Tw_n) is increasing, it is not un-convergent by Lemma 1.2(ii). Similarly, no subsequence of (Tw_n) is un-convergent. $\square$

We do not know whether the converse is true.

Here $T_n \uparrow T$ is understood in the usual operator order: the sequence is increasing and has supremum T ; in particular, $T - T_1$ is positive. A Banach lattice is a *KB-space* when every increasing norm-bounded sequence in its positive cone converges in norm. A net (x_α) un-converges to x when

$$\| |x_\alpha - x| \wedge u \| \longrightarrow 0 \quad (u \in X_+).$$

Proof intuition

The monotone operator sequence itself barely needs to be analyzed. Its first term already has the required sequential compactness, while the positive gap $T - T_1$ has domain c_0 . Positivity turns the vectors Se_k into a positive series whose partial sums are uniformly bounded: they are the images of the unit vectors $e_1 + \dots + e_N$ of c_0 . The KB property makes these partial sums converge in norm. Their tails then dominate the image of every unit-ball tail in c_0 , uniformly, so S is the operator-norm limit of its finite-rank coordinate truncations. Thus the whole positive gap is compact, and adding it to T_1 preserves sequential un-compactness.

The positive-gap lemma

Lemma 1. *If X is a KB-space, then every positive bounded operator $S : c_0 \rightarrow X$ is compact.*

Proof. Let (e_k) be the standard unit-vector basis of c_0 , let P_N be the projection onto the first N coordinates, and put

$$x_k = Se_k \in X_+, \quad s_N = \sum_{k=1}^N x_k = S(e_1 + \dots + e_N).$$

The sequence (s_N) is increasing and

$$\|s_N\| \leq \|S\| \|e_1 + \dots + e_N\|_\infty = \|S\|.$$

Since X is a KB-space, there is $s \in X_+$ such that $s_N \rightarrow s$ in norm. Moreover $s_N \leq s$ for every N : for fixed N , $s_m - s_N \in X_+$ when $m \geq N$, and the positive cone is norm closed.

Fix N and $a \in B_{c_0}$. For $m > N$, positivity of S gives

$$\begin{aligned} |SP_m(I - P_N)a| &\leq S|P_m(I - P_N)a| \\ &\leq \sum_{k=N+1}^m x_k = s_m - s_N \leq s - s_N. \end{aligned}$$

As $m \rightarrow \infty$, $P_m(I - P_N)a \rightarrow (I - P_N)a$ in c_0 . Continuity of S and of the lattice modulus, together with norm closedness of X_+ , therefore yields

$$|S(I - P_N)a| \leq s - s_N.$$

Consequently,

$$\|S(I - P_N)\| \leq \|s - s_N\| \longrightarrow 0.$$

Each SP_N has finite rank, so S is compact. □

Full characterization

Theorem 1. *For a Banach lattice X , the following are equivalent.*

1. X is a KB-space.
2. Whenever (T_n) is a sequence of sequentially un-compact operators in $L(c_0, X)$ and $T_n \uparrow T$, the operator T is sequentially un-compact.

Proof. The implication (2) $\Rightarrow$ (1) is Proposition 9.6 of [1].

Suppose (1), and let sequentially un-compact operators $T_n : c_0 \rightarrow X$ satisfy $T_n \uparrow T$. Then

$$S := T - T_1 \geq 0.$$

Lemma 1 shows that S is compact. In particular, S is sequentially un-compact, since every norm-convergent sequence is un-convergent.

The sequentially un-compact operators form a linear subspace [1, Proposition 9.2]. For completeness, if (z_j) is bounded, first pass to a subsequence along which $T_1 z_j$ un-converges, and then to a further subsequence along which $S z_j$ un-converges. Addition is continuous for un-convergence, so $(T_1 + S)z_j = T z_j$ un-converges along the final subsequence. Thus T is sequentially un-compact, proving (2). $\square$

Remark 1 (A slightly stronger form). *The proof does not use the intermediate operators $T_2, T_3, \dots$. If X is a KB-space, $R : c_0 \rightarrow X$ is sequentially un-compact, and $T - R \geq 0$, then T is sequentially un-compact. Indeed, the positive gap is compact by Lemma 1.*

Verification notes

The main possible pitfall is to write the non- c_0 vector $(1, 1, \dots)$ and apply S to it. The proof does not do this: s is constructed as a norm limit in X , and the tail estimate is first proved for finite truncations. Sign-changing unit-ball vectors are covered by the positivity inequality $|Sz| \leq S|z|$. Passing to the limit uses only norm continuity and the closed positive cone. Finally, the order hypothesis is used only in the valid implication $T_1 \leq T \Rightarrow T - T_1 \geq 0$; no norm convergence of T_n is assumed.

An independent line-by-line audit is included in `verification.md`. Its verdict is *likely valid, candidate full solution pending human review*.

Bounded novelty check and limitations

On August 9, 2026, the search covered the run’s registry, solution, attempt, and proof-gap indexes; exact web searches for the source wording and close variants involving “sequentially un-compact”, “order closed”, c_0 , and “KB-space”; and a local full-text scan of all 43 ingested arXiv source papers that cite the exact title of [1], through the 2026 corpus. No later paper stating or proving Theorem 1 was found.

The search did locate later resolutions of two *different* questions. The paper arXiv:2304.04189 proves norm-closedness of un-compact operators, while arXiv:2404.15641 answers the band-lifting question. This helps separate those literature-closed targets from the converse addressed here. The present novelty assessment is nevertheless bounded, not definitive: non-arXiv literature was not exhaustively citation-indexed, and the positive c_0 lemma is elementary enough that it may exist elsewhere in a different form.

Human-review recommendation

Review as a full solution. The decisive point is the uniform tail estimate in Lemma 1; once it is accepted, the missing implication is immediate. A reviewer should also check whether this elementary positive- operator observation has already been explicitly connected to Proposition 9.6 in non-arXiv literature.

References

- [1] M. Kandić, M. A. A. Marabeh, and V. G. Troitsky, *Unbounded norm topology in Banach lattices*, J. Math. Anal. Appl. **451** (2017), 259–279; arXiv:1608.05489; doi:10.1016/j.jmaa.2017.01.041.